

Apply filters to SQL queries

Project description

My organization wanted to make its system more secure. My role was to **investigate potential security issues** and **update employee computers** as needed. I used SQL queries with filters to identify, analyze, and take action on specific security-related scenarios.

Retrieve after hours failed login attempts

There was a failed login attempt that happened after business hours (after 6:00 PM). All after hours login attempts that failed need to be investigated. The following code sniffed how I created a SQL query to filter for failed login attempts that occurred.

```
MariaDB [organization]> SELECT *  
  -> FROM log_in_attempts  
  -> WHERE login_time > '18:00' AND success = FALSE ;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	astrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
69	wjaffrey	2022-05-11	19:55:15	USA	192.168.100.17	0
82	abernard	2022-05-12	23:38:46	MEX	192.168.234.49	0
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
104	asundara	2022-05-11	18:38:07	US	192.168.96.200	0
107	bisles	2022-05-12	20:25:57	USA	192.168.116.187	0
111	astrada	2022-05-10	22:00:26	MEXICO	192.168.76.27	0
127	abellmas	2022-05-09	21:20:51	CANADA	192.168.70.122	0
131	bisles	2022-05-09	20:03:55	US	192.168.113.171	0
155	cgriffin	2022-05-12	22:18:42	USA	192.168.236.176	0
160	jclark	2022-05-10	20:49:00	CANADA	192.168.214.49	0
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0

19 rows in set (0.001 sec)

The first part of the screenshot is my query, and the second part is a portion of the output. I used the WHERE clause with an AND operator to filter my results to output only login attempts that occurred after 18:00 and were unsuccessful.

Retrieve login attempts on specific dates

A suspicious event happened on 2022-05-09. To investigate, I needed to look at all login attempts that occurred on that date and also the day before 2022-05-08. The following code screenshot demonstrates how I created a SQL query to filter for failed login attempts that occurred after business hours.

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08' ;
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0

Retrieve login attempts outside of Mexico

While reviewing the organization's login activity, I noticed there might be an issue with login attempts that happened from outside of Mexico. Since the company only expects users to log in from Mexico, these attempts should be flagged and investigated. I wrote a SQL query to filter for all login attempts where the country is not Mexico.

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0

Retrieve employees in Marketing

My team needed to update the computers for certain employees in the Marketing department. To do this, I had to pull information on which employee machines belonged to marketing employees located in the East building. I wrote a SQL query that filters employee machine records by department and building.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Marketing' AND office LIKE 'East%';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267

Retrieve employees in Finance or Sales

After the Marketing department updates, my team also needed to apply a different security update to machines belonging to employees in the Finance and sales departments. To prepare for this, I created a SQL query to find all employee machines from only these two departments.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Finance' OR department = 'Sales' ;
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134

Retrieve all employees not in IT

My team needed to apply a security update to all employees except those in the information Technology(IT) department. To do this, I created a SQL to retrieve information on employees who are not in IT.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE NOT department = 'Information Technology' ;
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153

Summary

In my SQL practice, I worked on extracting specific information from two tables : login_attempts and employees. I applied different filters to get the exact data needed for each scenario.

Techniques I used:

- AND, OR, and NOT operators
- LIKE with % wildcard
- Date and time filters
- IN operator

Examples of tasks I performed:

- Identified **failed login attempts** after business hours or on specific dates.
- Investigated login attempts **outside expected locations**, such as outside Mexico.
- Retrieved **employee machine records** for different departments (Marketing, Finance, Sales) or excluding IT.
- Applied filters for **department, building, or specific patterns** to support IT security updates.